Dear Quillstone AI Hiring Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. My experience building and evaluating forecasting and customer-propensity models, developing Python and SQL machine-learning workflows, including feature engineering, scheduled batch scoring, and forecasting pipelines, and working closely with product and operations teams has given me a practical foundation in applied machine learning workflows and production-oriented analytical systems.

At Thistlebox, I built a weekly demand-forecasting workflow using LightGBM and engineered calendar, price and menu-position features. It runs as a scheduled Vertex AI Workbench notebook and writes forecasts to BigQuery; forecast WAPE fell from 23% to 14%, with an estimated 9% reduction in food waste. I also developed a weekly XGBoost churn-scoring workflow in BigQuery, where a 10% holdout validated a 2.1 percentage-point reduction in churn in the treated group. These projects have involved turning business needs into useful models and data workflows, evaluating their impact, and communicating results to stakeholders.

Alongside this work, I design and analyse around 40 A/B tests a year and built a Python Streamlit sequential-testing tool used by about 25 colleagues. Since 2025, I have written pytest unit tests for forecasting feature code, reaching approximately 60% coverage; I also wrote the Dockerfile for a Streamlit application used by around 25 colleagues. The data platform team manages its Cloud Run deployment. I use BigQuery daily for production SQL and batch churn scoring, maintain dbt models for the analytics layer, and have worked with scheduled Vertex AI Workbench notebook jobs. I am keen to build further on this experience in production machine learning engineering. My MSc in Statistics and BSc in Mathematics support my work in modelling, experimentation and feature engineering.

Quillstone's focus on helping merchants through its customer operations tools, and its emphasis on team ownership, practical problem-solving and learning, make this role particularly appealing to me. I would welcome the opportunity to discuss how my experience in applied modelling, experimentation and analytical workflows could contribute to the AI team through practical model evaluation, feature engineering, experimentation, and reliable data workflows while developing further in production machine learning engineering.

Kind regards, Tobias Mwangi-Hartley